

Minerva: Solving Quantitative Reasoning Problems with Language Models

Google Research

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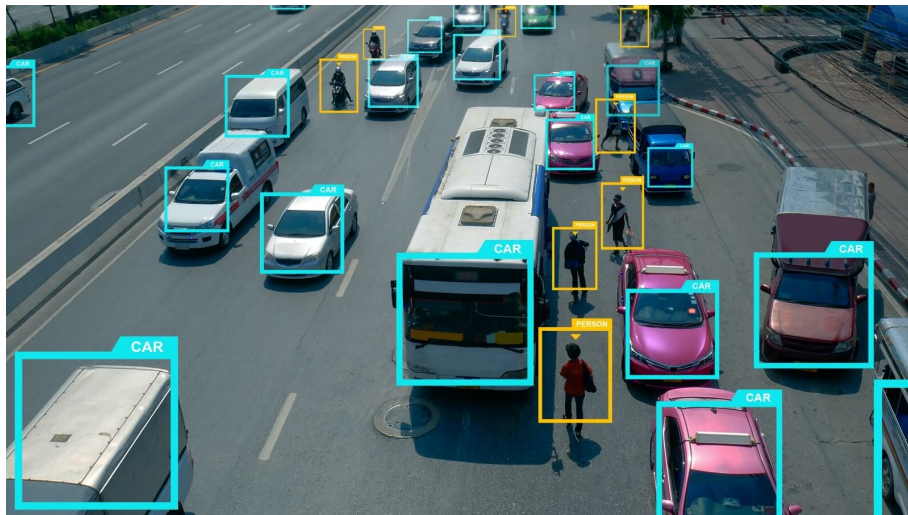
Conclusions



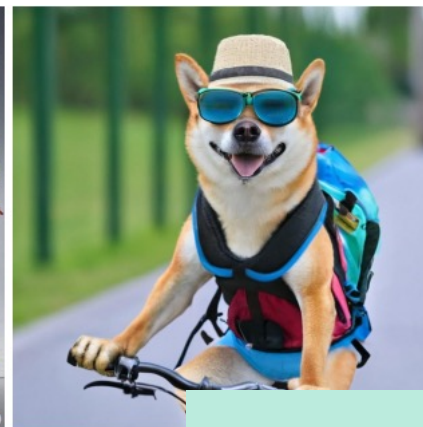
Quantitative Reasoning Problems

- 정량적 추론문제
 - 수학문제, 과학문제
 - 수치적인 연산이 포함되고, 정답이 정해져있는 문제
- Datasets:
 - Math Word Problem (MWP)
 - Grade School Math (GSM8k)
 - Massive Multitask Language Understanding (MMLU)

Modern model



Sprouts in the shape of text 'Imagen' coming out of a fairytale book.



A photo of a Shiba Inu on a bike. It is wearing sunglasses.



Teddy bears swimming at the Olympics 400m Butterfly event.



A cute corgi lives in Japan.



Quantitative Reasoning Problems

Question: A line parallel to $y = 4x + 6$ passes through $(5, 10)$. What is the y -coordinate of the point where this line crosses the y -axis?

Target: Since the line is parallel to $y = 4x + 6$, we know that its slope is 4. Thus the equation of the line is $y = 4x + b$ where b is the y -intercept. Plugging $(5, 10)$ into this gives $10 = 4 \cdot 5 + b \Rightarrow b = \boxed{-10}$ which is what we wanted.

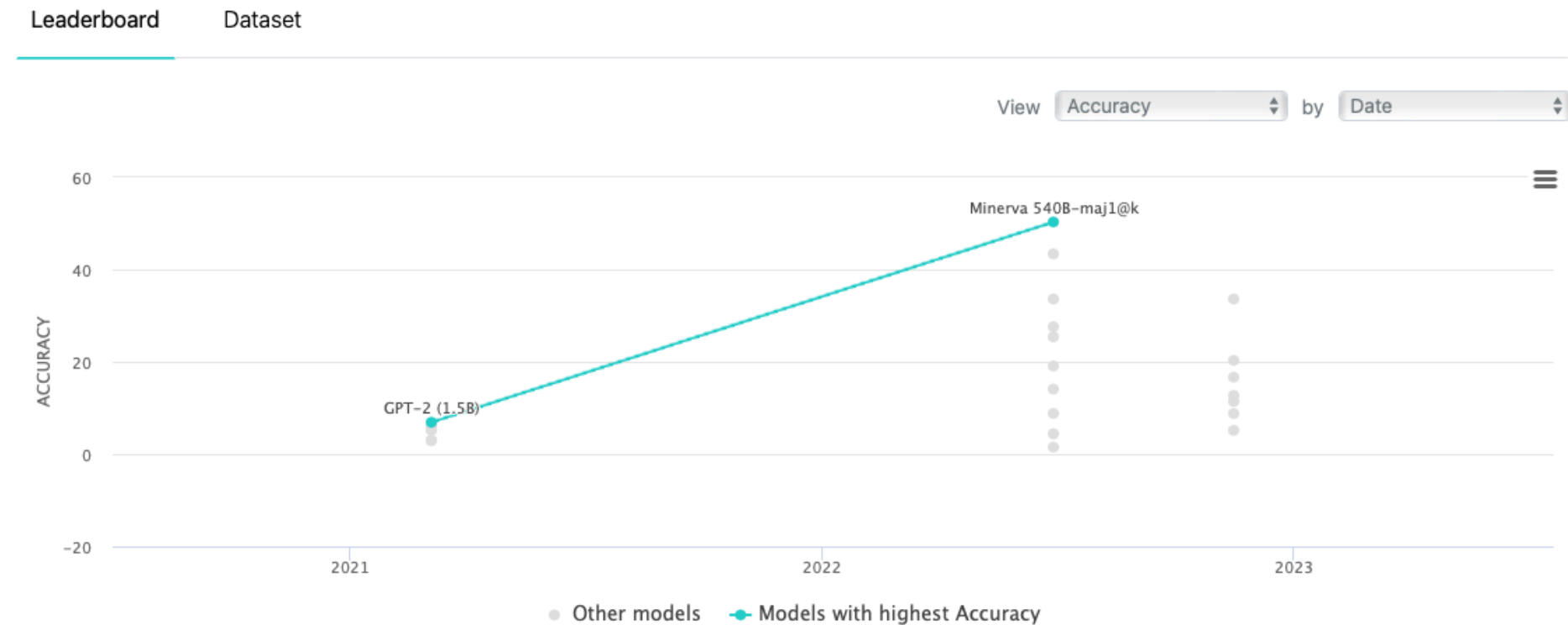
Model output:

Quantitative Reasoning Problems

- GPT, BERT로 접근.
- 언어모델이 양적 정보(숫자 등)를 잘 학습하지 못함.
- 외부 도구에 의존해야 할 수 있음.

Quantitative Reasoning Problems

Math Word Problem Solving on MATH



Quantitative Reasoning Problems

〈서술형 수학문제〉

한 변의 길이가 24cm인 정육각형과 둘레가 같은 정팔각형이 있습니다. 이 정팔각형의 한 변의 길이는 몇 cm 인지 구하시오.

인공지능
모델

〈풀이 과정〉

〈해답〉

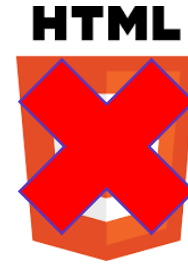
```
a = 24  
b = 6  
c = 8  
y = a * b // c  
print(y)
```

18

외부 도구

Minerva

- 학습 데이터:
 - Math Web Pages
 - arXiv
 - General Natural Language Data
- 학습 방법
 - few-shot prompting
 - chain of thought & scratchpad prompting
 - majority voting – nucleus sampling
 - based on the PaLM general language models



Minerva

Question: A line parallel to $y = 4x + 6$ passes through $(5, 10)$. What is the y -coordinate of the point where this line crosses the y -axis?

Target: Since the line is parallel to $y = 4x + 6$, we know that its slope is 4. Thus the equation of the line is $y = 4x + b$ where b is the y -intercept. Plugging $(5, 10)$ into this gives $10 = 4 \cdot 5 + b \Rightarrow b = \boxed{-10}$ which is what we wanted.

Model output:

Prompt

```
user@host:/home$ _
```

- 다음 대사를 상기시켜주기 위한 힌트

Thought of chain

Standard Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain of Thought Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

Scratchpad Prompt

SCRATCHPAD TRACING

Consider the following Python function:

```
def remove_Occ(s,ch):  
    for i in range(len(s)):  
        if (s[i] == ch):  
            s = s[0 : i] + s[i + 1:]  
            break  
    for i in range(len(s) - 1,-1,-1):  
        if (s[i] == ch):  
            s = s[0 : i] + s[i + 1:]  
            break  
    return s
```

```
output = remove_Occ("PHP","P")
```

What is the execution trace?

Large Language
Model

```
state: {}  
line: def remove_Occ(s,ch):  
state: {"remove_Occ": "<callable_object remove_Occ>"}  
line: output = remove_Occ("PHP","P")  
state: {"ch": "P", "s": "PHP"}  
line:     for i in range(len(s)):  
state: {"ch": "P", "s": "PHP", "i": 0}  
line:         if (s[i] == ch):  
state: {"ch": "P", "s": "PHP", "i": 0}  
line:             s = s[0 : i] + s[i + 1:]  
state: {"ch": "P", "s": "HP", "i": 0}  
line:             break  
state: {"ch": "P", "s": "HP", "i": 0}  
line:     for i in range(len(s) - 1,-1,-1):  
state: {"ch": "P", "s": "HP", "i": 1}  
line:         if (s[i] == ch):  
state: {"ch": "P", "s": "HP", "i": 1}  
line:             s = s[0 : i] + s[i + 1:]  
state: {"ch": "P", "s": "H", "i": 1}  
line:             break  
state: {"ch": "P", "s": "H", "i": 1}  
line:     return s  
state: {"remove_Occ": "<callable_object remove_Occ>",  
      "output": "H"}
```

Minerva Prompt

```
1 Problem:
2 Find the domain of the expression  $\frac{\sqrt{x-2}}{\sqrt{5-x}}$ .
3
4 Solution:
5 The expressions inside each square root must be non-negative. Therefore,
6  $x-2 \geq 0$ , so  $x \geq 2$ , and  $5-x \geq 0$ , so  $x \leq 5$ . Also, the denominator
7 cannot be equal to zero, so  $5-x > 0$ , which gives  $x < 5$ . Therefore, the domain of
8 the expression is  $\boxed{[2,5)}$ .
9 Final Answer: The final answer is  $[2,5)$ . I hope it is correct.
10
11 Problem:
12 If  $\det \mathbf{A} = 2$  and  $\det \mathbf{B} = 12$ , then find
13  $\det (\mathbf{A} \mathbf{B})$ .
14
15 Solution:
16 We have that  $\det (\mathbf{A} \mathbf{B}) = (\det \mathbf{A})(\det \mathbf{B})$ 
17  $= (2)(12) = \boxed{24}$ .
18 Final Answer: The final answer is  $24$ . I hope it is correct.
19
20 Problem:
21 Terrell usually lifts two 20-pound weights 12 times. If he uses two 15-pound
22 weights instead, how many times must Terrell lift them in order to lift the
23 same total weight?
24
```

Majority voting

- 256개 이상의 output sampling. Nucleus sampling
- 나올 수 있는 정답이 제한되지 않기 때문에 유효함.

Minerva

Majority voting

- 256개 이상의 output sampling. Nucleus sampling
- 나올 수 있는 정답이 제한되지 않기 때문에 유효함.

```
import torch
import torch.nn.functional as F

a = torch.tensor([1,2,3,4])
F.softmax(a, dim=0) # tensor([0.0321, 0.0871, 0.2369, 0.6439])
F.softmax(a/.5, dim=0) # tensor([0.0021, 0.0158, 0.1171, 0.8650])
F.softmax(a/1.5, dim=0) # tensor([0.0708, 0.1378, 0.2685, 0.5229])
F.softmax(a/1e-6, dim=0) # tensor([0., 0., 0., 1.])
```

Question: A particle moves so that it is at $(3 \sin \frac{t}{4}, 3 \cos \frac{t}{4})$ at time t . Find the speed of the particle, measured in unit of distance per unit of time.

Target: The parametric curve $(x, y) = (3 \sin \frac{t}{4}, 3 \cos \frac{t}{4})$ describes a circle with radius 3. Furthermore, it makes a full revolution at time $t = 8\pi$. (figure) Therefore, the speed of the particle is $\frac{6\pi}{8\pi} = \boxed{\frac{3}{4}}$.

Model output: The speed is the magnitude of the velocity vector. Therefore, we must first find the velocity vector, which is the derivative of the position vector. We have

$$\mathbf{v} = \frac{d}{dt} \left(3 \sin \frac{t}{4}, 3 \cos \frac{t}{4} \right) = 3 \left(\cos \frac{t}{4}, -\sin \frac{t}{4} \right) \frac{1}{4} = \left(\frac{3}{4} \cos \frac{t}{4}, -\frac{3}{4} \sin \frac{t}{4} \right).$$

Therefore, the magnitude of the velocity vector is

$$\begin{aligned} |\mathbf{v}| &= \sqrt{\left(\frac{3}{4} \cos \frac{t}{4} \right)^2 + \left(-\frac{3}{4} \sin \frac{t}{4} \right)^2} = \sqrt{\frac{9}{16} \cos^2 \frac{t}{4} + \frac{9}{16} \sin^2 \frac{t}{4}} \\ &= \frac{3}{4} \sqrt{\cos^2 \frac{t}{4} + \sin^2 \frac{t}{4}} = \frac{3}{4}. \end{aligned}$$

Therefore, the speed of the particle is $\boxed{\frac{3}{4}}$.

Conclusions

	MATH	OCWCourses	GSM8k	MMLU-STEM
PaLM 8B	1.5%	1.5%	4.1%	22.0%
Minerva 8B	14.1%	7.7%	16.2%	35.6%
Minerva 8B, maj1@k	25.4%	12.5%	28.4%	43.4%
PaLM 62B	4.4%	5.9%	33.0%	39.1%
Minerva 62B	27.6%	12.9%	52.4%	53.9%
Minerva 62B, maj1@k	43.4%	23.5%	68.5%	63.5%
PaLM 540B	8.8%	7.1%	56.5%	58.7%
Minerva 540B	33.6%	17.6%	58.8%	63.9%
Minerva 540B, maj1@k	50.3%	30.8%	78.5%	75.0%
OpenAI davinci-002	19.1%	14.8%	-	-
Published SOTA	6.9% ^a	-	74.4% ^b	54.9% ^c

Conclusions

- 고품질 데이터 셋을 확보함으로써 외부 도구의 도움 없이 정량적 문제를 효과적으로 해결할 수 있다.
- 풀이과정까지 맞았는지 확인할 방법은 없다.
- 코드 생성 모델과 결합할 경우 좋은 성능을 낼 여지가 있다.

MINERVA

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THANK YOU